

Geospatial Flood-Risk Modelling using Climate Data and Hidden Markov Models in Rwanda

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The problem

Kigali floods after intense rain, yet official hazard maps are static: they show where flooding is possible, not when or how widely flood pressure rises day to day after rainfall.

What I built

- A 250 m grid (729 cells) over the corridor; one record per cell per day, 2018–2024 (1.86M records).
- Real, free, keyless data: rainfall (Open-Meteo ERA5), elevation+slope (SRTM), rivers/roads/buildings (OpenStreetMap), and the official flood polygons (Rwanda GeoPortal / Ministry of Environment).
- Models: rainfall-threshold & static-polygon baselines, Logistic Regression, Decision Tree, Random Forest, XGBoost (+SHAP), and a Hidden Markov Model for daily state transitions.
- Evaluated on a strict temporal hold-out: train 2018–2023, test on the unseen year 2024.

Headline results (2024 hold-out)

- Random Forest: macro-F1 0.831, high-pressure recall 0.867 (targets were 0.75 and 0.80). XGBoost close behind (0.813 / 0.847).
- Baselines clearly beaten (rainfall-threshold 0.62, static-polygon 0.32) — geospatial context adds real value.
- Spatial validation vs official polygons: enrichment 1.59×, inside-vs-outside high odds 1.84× — without ever training on the polygons.

Contribution

A reproducible, interpretable, time-aware flood-pressure tool that independently reinforces the official flood map and extends it with a dynamic, rainfall-driven footprint — delivered with a Streamlit dashboard for inspection by date and location.

Note: a flood-pressure state is a derived risk condition, not a confirmed flood event; the tool complements, not replaces, official early warning.